



Software Engineering Fundamentals Project Phase Four (Prototype, Testing, and Reflection)

Group 4

TeamNames – ID :

- **Ghala Al-Shamrani –**
 - **Dana Al-Bushri –**
 - **Danah Bukhari –**
 - **Ghala Al-Hazmi –**
 - **Huda Al-Halaf –**
- **Najd Al-Thubaity –**

Dr. Basmah M. Alsenani

Prototype

We created a high-fidelity prototype for SkillSwap. The prototype includes the login page, Sign up page, home page, skill Request page, messaging page, and user profile page. Users can browse skills, communicate with others, and exchange skills easily through the system.

1/ Login Screen



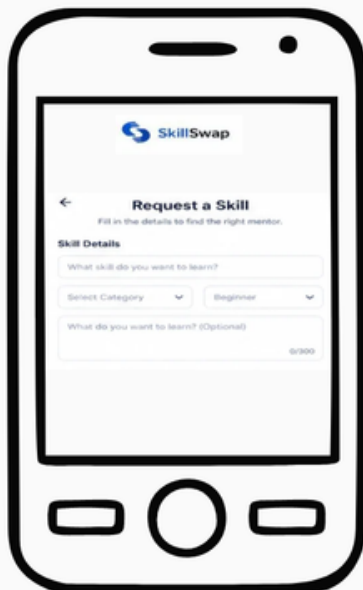
2/ Sign Up Screen



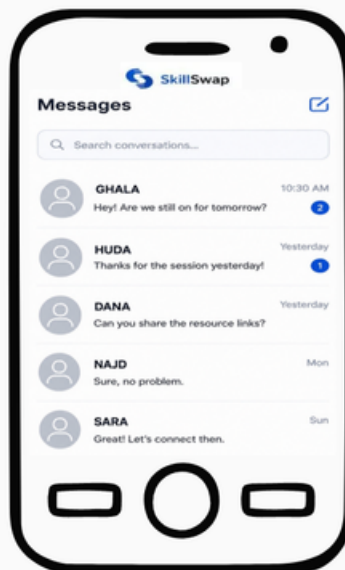
3/ Home Screen



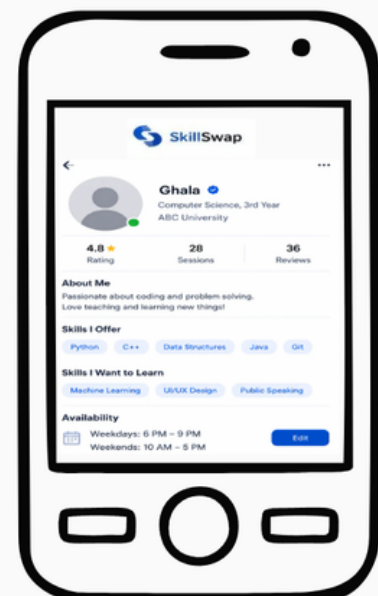
4/ Skill Request Screen



5/ Messages Screen



6/ User profile Screen



the clickable prototype link : [Prototype for Skill swap](#)

Test Cases

Test Case 1: User Registration

Action: Create a new account using valid information.

Expected Result: Account is created successfully and user is redirected to the home page.

Test Case 2: User Login

Action: Login using a registered email and password.

Expected Result: User successfully accesses the system.

Test Case 3: Add a Skill

Action: User adds a new skill to their profile.

Expected Result: The skill appears in the user's profile and becomes available for matching.

Test Case 4: Send Skill Exchange Request

Action: User sends a request to another user for a skill exchange session.

Expected Result: The request is delivered and recorded in the system.

Test Case 5: Send a Message

Action: User sends a message through the chat feature.

Expected Result: The message is successfully delivered to the recipient.



Testing Results

The system performed as expected during testing. All major functions were tested successfully, including account creation, login, skill management, request submission, and messaging.

No critical errors were encountered during testing.

Minor interface improvements were identified, such as enhancing navigation clarity and improving visual consistency between screens.

Overall, the prototype achieved its intended objectives and demonstrated the core functionality of the SkillSwap platform.



System Limitations

Although the prototype successfully demonstrates the core features of SkillSwap, it has some limitations:

- The system does not support real-time notifications.
- Voice and video communication features are not available.
- The matching process is based on user-entered skills and does not use advanced AI algorithms.
- The system currently supports a limited number of users for testing purposes.
- External integrations with university systems are not included.



Reflection

Through this project, we gained valuable experience in software engineering concepts, including requirements analysis, system design, prototyping, and testing. We learned how to transform an idea into a structured software solution while following the software development lifecycle.

One of the main challenges was designing a system that effectively connects users based on their skills while keeping the interface simple and user-friendly. Another challenge was organizing system requirements and creating appropriate diagrams and documentation.

In the future, we would like to improve the system by adding real time notifications, AI based skill matching, video communication features, and stronger security measures. These enhancements would improve the overall user experience and increase the effectiveness of the platform.



Individual Contribution

Each team member contributed to the project as follows:

Ghala Al-Hazmi

- Prototype

Ghala Al-Shamrani

- Test Cases

Dana Al-Bushri

- Testing Results

Danah Bukhari

- System Limitations

Najd Al-Thubaity

- Reflection

Huda Al-Halaf

- Formatting and final review



جامعة أم القرى
UMM AL-QURA UNIVERSITY



جامعة أم القرى
UMM AL-QURA UNIVERSITY

